화학수학

열한 번째 수업

미분방정식

Homogeneous Linear ODEs with Const. Coeff.

$$a\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + b\frac{\mathrm{d}y}{\mathrm{d}x} + cy = 0$$

- Trial solution (시행해): $y = e^{\lambda x}$

$$a\lambda^2 e^{\lambda x} + b\lambda e^{\lambda x} + ce^{\lambda x} = 0$$

- Characteristic equation (고유방정식)

$$a\lambda^2 + b\lambda + c = 0$$

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Roots of Characteristic Equation

$$a\lambda^2 + b\lambda + c = 0$$

$$\lambda_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \qquad \lambda_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

1. Two roots are real and distinct ($b^2 - 4ac > 0$)
2. Two roots are real and equal ($b^2 - 4ac = 0$)
3. Two roots are complex numbers ($b^2 - 4ac < 0$)

(1) Distinct Real Roots

$$\lambda_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \qquad \lambda_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

We find two solutions: $y_1 = e^{\lambda_1 x}$ and $y_2 = e^{\lambda_2 x}$

The general solution is

$$y = c_1 y_1 + c_2 y_2 = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

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(2) Repeated Real Roots

$$\lambda_1 = \lambda_2 = \frac{-b}{2a}$$

We obtain one solution: $y_1 = e^{\lambda_1 x}$

Using the reduction of order,

$$y_2 = y_1 \int \frac{e^{-\int P \mathrm{d}x}}{y_1^2} \mathrm{d}x = e^{\lambda_1 x} \int \frac{e^{2\lambda_1 x}}{e^{2\lambda_1 x}} \mathrm{d}x = x e^{\lambda_1 x}$$

$$y = c_1 y_1 + c_2 y_2 = c_1 e^{\lambda_1 x} + c_2 x e^{\lambda_1 x}$$

(3) Conjugate Complex Roots

$$\lambda_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} = \alpha + i\beta$$

$$\lambda_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a} = \alpha - i\beta$$

We find two solutions: $y_1 = e^{\lambda_1 x}$ and $y_2 = e^{\lambda_2 x}$

The general solution is

$$y = e^{\alpha x}(c_1' \cos \beta x + c_2' \sin \beta x)$$

Application: Harmonic Oscillator (조화진동자)

- Newton's second law

$$F = ma = m\frac{\mathrm{d}v}{\mathrm{d}t} = m\frac{\mathrm{d}^2x}{\mathrm{d}t^2}$$

- Equation of motion

$$F = m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -kx$$

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \frac{k}{m}x = 0$$

Solving Equation of Motion

$$\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} + \frac{k}{m}x = 0$$

$$\lambda_1 = i\sqrt{k/m}, \qquad \lambda_2 = -i\sqrt{k/m}$$

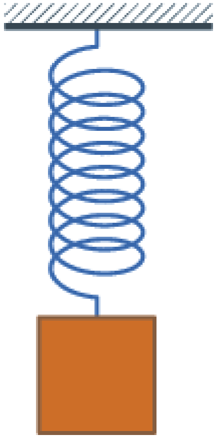
Define $\omega = \sqrt{k/m}$; the general solution is

$$x(t) = c_1 \cos \omega t + c_2 \sin \omega t$$

Damped Harmonic Oscillator

(감쇠 조화진동자)

- Damping (감쇠): reduction of oscillation
- Application of friction force (마찰력)
  - proportional to the velocity
  - in the opposite direction



$$\vec{F}_f = -\zeta\vec{v} = -\zeta\frac{\mathrm{d}\vec{r}}{\mathrm{d}t}$$

friction constant
(마찰 상수)

(Image: https://commons.wikimedia.org/wiki/File:Damped_spring.gif)

1-Dimensional Equation of Motion

$$F = m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -kx - \zeta\frac{\mathrm{d}x}{\mathrm{d}t}$$

- Standard form

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \frac{\zeta}{m}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{k}{m}x = 0$$

- Characteristic equation

$$\lambda^2 + \frac{\zeta}{m}\lambda + \frac{k}{m} = 0$$

Roots of Characteristic Equation

$$\lambda^2 + \frac{\zeta}{m}\lambda + \frac{k}{m} = 0$$

$$D = \left(\frac{\zeta}{m}\right)^2 - \frac{4k}{m}$$

1. Distinct real roots ($D > 0$) → overdamping
2. Repeated real roots ($D = 0$) → critical damping
3. Conjugate complex roots ($D < 0$) → underdamping

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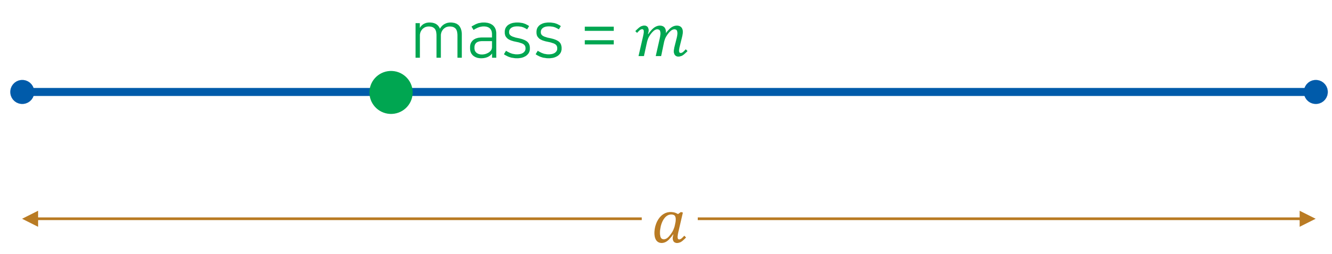
Application: Particle in a Box



$$\frac{\mathrm{d}^2\psi(x)}{\mathrm{d}x^2} + \frac{2mE}{\hbar^2}\psi(x) = 0 \quad (E > 0)$$

Boundary conditions: $\psi(0) = \psi(a) = 0$

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General Solution

$$\frac{\mathrm{d}^2\psi(x)}{\mathrm{d}x^2} + \frac{2mE}{\hbar^2}\psi(x) = 0 \quad (E > 0)$$

$$\lambda_1 = \frac{i\sqrt{2mE}}{\hbar}, \qquad \lambda_2 = -\frac{i\sqrt{2mE}}{\hbar}$$

Hence, the general solution is

$$\psi(x) = c_1 \cos\left(\frac{\sqrt{2mE}}{\hbar}x\right) + c_2 \sin\left(\frac{\sqrt{2mE}}{\hbar}x\right)$$

Inhomogeneous Linear ODE

$$a\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + b\frac{\mathrm{d}y}{\mathrm{d}x} + cy = \underline{g(x)}$$

inhomogeneous term
(비동차항)

- Complementary equation (보상방정식)

$$a\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + b\frac{\mathrm{d}y}{\mathrm{d}x} + cy = 0$$

Solving Inhomogeneous ODE

$$a\frac{\mathrm{d}^2y}{\mathrm{d}x^2}+b\frac{\mathrm{d}y}{\mathrm{d}x}+cy=g(x)$$

1. Solve the complementary equation →
   complementary function (보상함수) y_{c}
2. Find a particular solution y_{p} that satisfies the DE
3. The general solution is $y=y_{\mathrm{c}}+y_{\mathrm{p}}$

Trial Solutions for y_{p}

Inhomogeneous term	Trial solution
1	A
t^n	$A_0 + A_1 t + A_2 t + \cdots + A_n t^n$
$e^{\alpha t}$	$Ae^{\alpha t}$
$t^n e^{\alpha t}$	$e^{\alpha t}(A_0 + A_1 t + A_2 t + \cdots + A_n t^n)$
$e^{\alpha t} \sin(\beta t)$	$e^{\alpha t}[A \cos(\beta t) + B \sin(\beta t)]$
$e^{\alpha t} \cos(\beta t)$	$e^{\alpha t}[A \cos(\beta t) + B \sin(\beta t)]$

(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Table 12.1)

Glitch in the Method

A trial solution y_{p} can be

a solution of the complementary equation.

In this case, try $x^n y_{\mathrm{p}}$ instead!

(n: smallest integer that resolves the problem)

오늘의 수업

- Heterogeneous linear ODEs with constant coefficients
    - Application: harmonic oscillator
- Use of Laplace transform
- Partial differential equations
    - Separation of variables
    - Wave equation
    - Schrödinger equation

Application: Forced Harmonic Oscillator

$$F = m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -kx + \underbrace{F(t)}_{\text{external force}}$$

- Standard form

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \frac{k}{m}x = \frac{F(t)}{m}$$

- Complementary equation

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \frac{k}{m}x = 0$$

$$\rightarrow x_\mathrm{c}(t) = c_1 \cos \omega t + c_2 \sin \omega t$$

Example 12.8

Assume that the external force on a forced harmonic oscillator is given by

$$F(t) = F_0 \sin \alpha t$$

where F_0 and α are constants. Find the general solution to the equation of motion.

The equation of motion is

$$\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} + \frac{k}{m} x = \frac{F(t)}{m} = \frac{F_0}{m} \sin \alpha t$$

The trial function for $\sin \alpha t$ is $A \sin \alpha t + B \cos \alpha t$. Substitution gives

$$-A\alpha^2 \sin \alpha t - B\alpha^2 \cos \alpha t + \frac{k}{m}(A \sin \alpha t + B \cos \alpha t) = \frac{F_0}{m} \sin \alpha t$$

Example 12.8

Assume that the external force on a forced harmonic oscillator is given by

$$F(t) = F_0 \sin \alpha t$$

where F_0 and α are constants. Find the general solution to the equation of motion.

$$-A\alpha^2 \sin \alpha t - B\alpha^2 \cos \alpha t + \frac{k}{m}(A \sin \alpha t + B \cos \alpha t) = \frac{F_0}{m} \sin \alpha t$$

Rearranging the equation,

$$\left(-\alpha^2 + \frac{k}{m}\right) A \sin \alpha t + \left(-\alpha^2 + \frac{k}{m}\right) B \cos \alpha t = \frac{F_0}{m} \sin \alpha t$$

Hence, $B = 0$ and $A = \frac{F_0/m}{-\alpha^2+k/m} = \frac{F_0}{k-m\alpha^2}$.

Example 12.8

Assume that the external force on a forced harmonic oscillator is given by

$$F(t) = F_0 \sin \alpha t$$

where F_0 and α are constants. Find the general solution to the equation of motion.

$$x_p = \frac{F_0}{k - m\alpha^2} \sin \alpha t$$

As we know $x_{\mathrm{c}}(t) = c_1 \cos \omega t + c_2 \sin \omega t$, the general solution is

$$x(t) = c_1 \cos \omega t + c_2 \sin \omega t + \frac{F_0}{k - m\alpha^2} \sin \alpha t$$

Use of Laplace Transforms

- (Generalized) derivative theorem

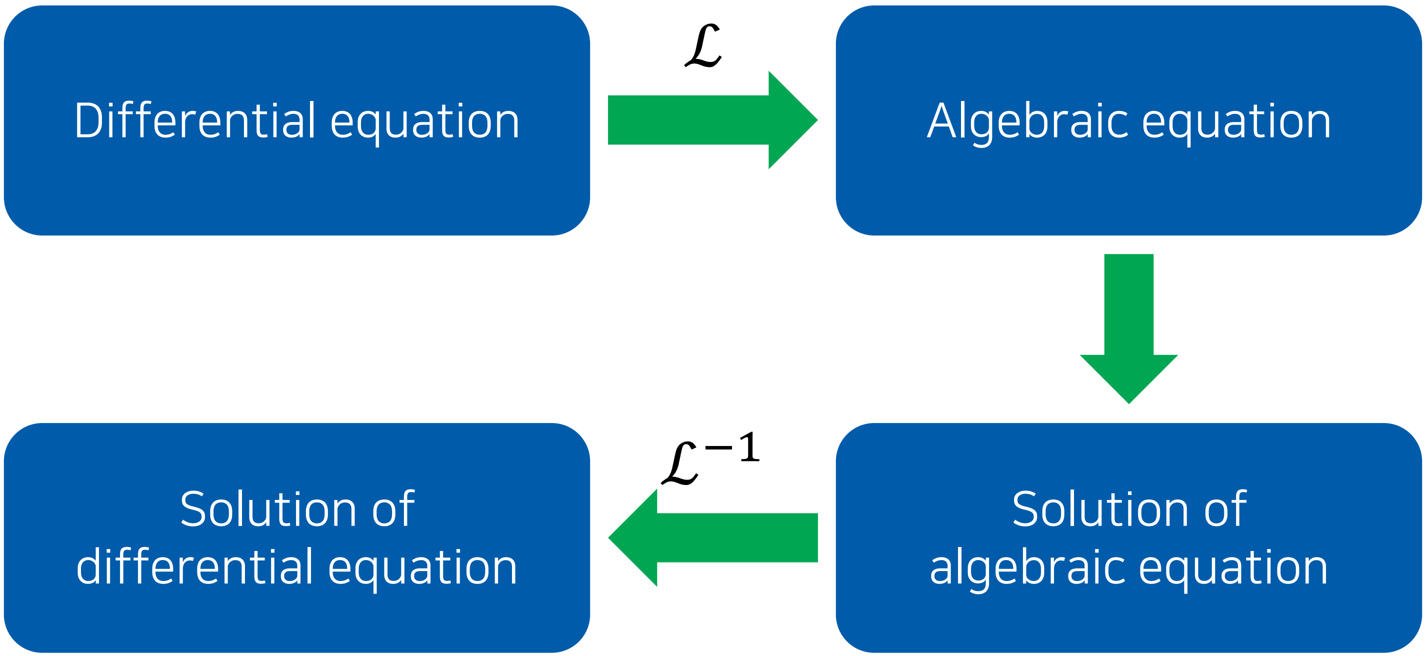
$$\mathcal{L}\left\{\frac{\mathrm{d}f(t)}{\mathrm{d}t}\right\} = s\mathcal{L}\{f(t)\} - f(0)$$

$$\mathcal{L}\left\{\frac{\mathrm{d}^2 f(t)}{\mathrm{d}t^2}\right\} = s^2\mathcal{L}\{f(t)\} - sf(0) - f'(0)$$

$$\mathcal{L}\{f^{(n)}(t)\} = s^n\mathcal{L}\{f(t)\} - s^{n-1}f(0) - s^{n-2}f'(0) - \cdots - f^{(n-1)}(0)$$

differential equation → algebraic equation

Solving DE with Laplace Transforms



Some Common Laplace Transforms

$F(s)$	$f(t)$
$1/s$	1
$1/s^2$	t
$n!/s^{n+1}$	t^n
$\dfrac{1}{s-a}$	$e^{at}\,(s \succ a)$
$\dfrac{s}{s^2+a^2}$	$\cos(at)$
$\dfrac{a}{s^2+a^2}$	$\sin(at)$
$\dfrac{s}{s^2-a^2}$	$\cosh(at)$
$\dfrac{a}{s^2-a^2}$	$\sinh(at)$

(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Table 11.1)

Example

Use the Laplace transform to solve the initial-value problem

$$\frac{\mathrm{d}x}{\mathrm{d}t} + 3x = 13\sin 2t\,, \qquad x(0) = 6.$$

Take the Laplace transform on each term:

$$\mathcal{L}\left\{\frac{\mathrm{d}x}{\mathrm{d}t}\right\} + 3\mathcal{L}\{x\} = 13\mathcal{L}\{\sin 2t\}$$

Use the known transforms:

$$[s\mathcal{L}\{x\} - x(0)] + 3\mathcal{L}\{x\} = \frac{26}{s^2+4}$$

$$(s+3)\mathcal{L}\{x\} = \frac{26}{s^2+4} + 6$$

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Example

Use the Laplace transform to solve the initial-value problem

$$\frac{\mathrm{d}x}{\mathrm{d}t} + 3x = 13\sin 2t\,, \qquad x(0) = 6.$$

$$(s+3)\mathcal{L}\{x\} = \frac{26}{s^2+4} + 6$$

Thus,

$$\mathcal{L}\{x\} = \frac{26}{(s+3)(s^2+4)} + \frac{6}{s+3}$$

$$\mathcal{L}\{x\} = \frac{6s^2+50}{(s+3)(s^2+4)}$$

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Example

Use the Laplace transform to solve the initial-value problem

$$\frac{\mathrm{d}x}{\mathrm{d}t} + 3x = 13\sin 2t\,, \qquad x(0) = 6.$$

$$\mathcal{L}\{x\} = \frac{6s^2+50}{(s+3)(s^2+4)}$$

Applying the method of partial fractions,

$$\frac{6s^2+50}{(s+3)(s^2+4)} = \frac{A}{s+3} + \frac{Bs+C}{s^2+4} = \frac{A(s^2+4)+(Bs+C)(s+3)}{(s+3)(s^2+4)}$$

As $A = 8,\ B = -2,\ C = 6,$

$$\mathcal{L}\{x\} = \frac{8}{s+3} + \frac{-2s}{s^2+4} + \frac{6}{s^2+4}$$

Example

Use the Laplace transform to solve the initial-value problem

$$\frac{\mathrm{d}x}{\mathrm{d}t} + 3x = 13\sin 2t\,, \qquad x(0) = 6.$$

$$\mathcal{L}\{x\} = \frac{8}{s+3} + \frac{-2s}{s^2+4} + \frac{6}{s^2+4}$$

Hence,

$$\mathcal{L}\{x\} = 8\mathcal{L}\{e^{-3t}\} - 2\mathcal{L}\{\cos 2t\} + 3\mathcal{L}\{\sin 2t\}$$

and

$$x(t) = 8e^{-3t} - 2\cos 2t + 3\sin 2t$$

Application: Underdamped Oscillator

$$\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} + \frac{\zeta}{m}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{k}{m}x = 0$$

Apply the Laplace transform:

$$[s^2\mathcal{L}\{x\} - sx(0) - x'(0)] + \frac{\zeta}{m}[s\mathcal{L}\{x\} - x(0)] + \frac{k}{m}\mathcal{L}\{x\} = 0$$

Solving the algebraic equation,

$$\mathcal{L}\{x\} = \frac{sx(0) + x'(0) + (\zeta/m)x(0)}{s^2 + (\zeta/m)s + (k/m)}$$

Rearrangement

$$\mathcal{L}\{x\} = \frac{sx(0) + x'(0) + (\zeta/m)x(0)}{s^2 + (\zeta/m)s + (k/m)}$$

$$= \frac{x(0)(s + \zeta/m) + x'(0)}{(s + \zeta/2m)^2 - \zeta^2/4m^2 + k/m}$$

Let $a = \zeta/2m$ and $\omega^2 = k/m - \zeta^2/4m^2 > 0$ (underdamping).

$$\mathcal{L}\{x\} = \frac{x(0)(s + 2a) + x'(0)}{(s + a)^2 + \omega^2}$$

$$= \frac{x(0)(s + a)}{(s + a)^2 + \omega^2} + \frac{ax(0) + x'(0)}{(s + a)^2 + \omega^2}$$

Known Facts

$$\mathcal{L}\{x\} = \frac{x(0)(s+a)}{(s+a)^2+\omega^2} + \frac{ax(0)+x'(0)}{(s+a)^2+\omega^2}$$

- Shifting theorem

$$\mathcal{L}\{e^{-at}f(t)\} = F(s+a)$$

- Known transforms

$$\mathcal{L}\{\cos\omega t\} = \frac{s}{s^2+\omega^2}, \qquad \mathcal{L}\{\sin\omega t\} = \frac{\omega}{s^2+\omega^2}$$

$$\mathcal{L}\{x\} = x(0)\mathcal{L}\{e^{-at}\cos\omega t\} + \frac{ax(0)+x'(0)}{\omega}\mathcal{L}\{e^{-at}\sin\omega t\}$$

Final Solution

$$\mathcal{L}\{x\} = x(0)\mathcal{L}\{e^{-at}\cos\omega t\} + \frac{ax(0) + x'(0)}{\omega}\mathcal{L}\{e^{-at}\sin\omega t\}$$

Hence, the inverse transform gives

$$x(t) = x(0)e^{-at}\cos\omega t + \frac{ax(0) + x'(0)}{\omega}e^{-at}\sin\omega t$$

$$x(t) = e^{-\zeta t/2m}\left[\underbrace{x(0)}_{c_1}\cos\omega t + \underbrace{\frac{ax(0) + x'(0)}{\omega}}_{c_2}\sin\omega t\right]$$

Partial Differential Equations

- General form of a linear second-order PDE

$$A\frac{\partial^2 u}{\partial x^2} + B\frac{\partial^2 u}{\partial x \partial y} + C\frac{\partial^2 u}{\partial y^2} + D\frac{\partial u}{\partial x} + E\frac{\partial u}{\partial y} + Fu = G$$

where $A, B, \cdots, G$ are functions of x and y.

- $G(x, y) = 0$: homogeneous (동차)
- $G(x, y) \neq 0$: inhomogeneous (비동차)

Separation of Variables

$$A\frac{\partial^2 u}{\partial x^2} + B\frac{\partial^2 u}{\partial x \partial y} + C\frac{\partial^2 u}{\partial y^2} + D\frac{\partial u}{\partial x} + E\frac{\partial u}{\partial y} + Fu = G$$

$X' = \mathrm{d}X/\mathrm{d}x\,, X'' = \mathrm{d}^2X/\mathrm{d}x^2, \cdots$
$Y' = \mathrm{d}Y/\mathrm{d}y\,, Y'' = \mathrm{d}^2Y/\mathrm{d}y^2, \cdots$

Assume that $u(x,y) = X(x)Y(y)$.

If we obtain $f(x, X, X', \cdots) = g(y, Y, Y', \cdots)$, let

$$f(x, X, X', \cdots) = g(y, Y, Y', \cdots) = \lambda$$

where λ is the separation constant.

(Tip: it is often convenient to use $\lambda = \kappa^2$ or $-\kappa^2$)

Example

Find solutions of

$$\frac{\partial^2 u}{\partial x^2} = 4\frac{\partial u}{\partial y}.$$

Assume $u(x, y) = X(x)Y(y)$ and obtain

$$X''Y = 4XY'$$

$$\frac{X''}{4X} = \frac{Y'}{Y}$$

Each side must be a constant. Consider the following three cases:

1) Separation constant > 0
2) Separation constant < 0
3) Separation constant = 0

Example

Find solutions of

$$\frac{\partial^2 u}{\partial x^2} = 4\frac{\partial u}{\partial y}.$$

1) Case 1:

$$\frac{X''}{4X} = \frac{Y'}{Y} = \kappa^2$$

Hence,

$$X'' - 4\kappa^2 X = 0,\ Y' - \kappa^2 Y = 0$$

which have the solutions

$$X(x) = c_1 e^{2\kappa x} + c_2 e^{-2\kappa x},\ Y(y) = c_3 e^{\kappa^2 y}$$

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Example

Find solutions of

$$\frac{\partial^2 u}{\partial x^2} = 4\frac{\partial u}{\partial y}.$$

1) Case 1:

$$X(x) = c_1 e^{2\kappa x} + c_2 e^{-2\kappa x},\ Y(y) = c_3 e^{\kappa^2 y}$$

The final solution is

$$u(x,y) = X(x)Y(y) = (c_1 e^{2\kappa x} + c_2 e^{-2\kappa x})c_3 e^{\kappa^2 y}$$

or

$$u(x,y) = A_1 e^{2\kappa x + \kappa^2 y} + B_1 e^{-2\kappa x + \kappa^2 y}$$

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Example

Find solutions of

$$\frac{\partial^2 u}{\partial x^2} = 4\frac{\partial u}{\partial y}.$$

2) Case 2:

$$\frac{X''}{4X} = \frac{Y'}{Y} = -\kappa^2$$

Hence,

$$X'' + 4\kappa^2 X = 0,\ Y' + \kappa^2 Y = 0$$

which have the solutions

$$X(x) = c_4 \cos 2\kappa x + c_5 \sin 2\kappa x\,,\ Y(y) = c_6 e^{-\kappa^2 y}$$

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Example

Find solutions of

$$\frac{\partial^2 u}{\partial x^2} = 4\frac{\partial u}{\partial y}.$$

2) Case 2:

$$X(x) = c_4 \cos 2\kappa x + c_5 \sin 2\kappa x\,,\;\; Y(y) = c_6 e^{-\kappa^2 y}$$

The final solution is

$$u(x, y) = X(x)Y(y) = (c_4 \cos 2\kappa x + c_5 \sin 2\kappa x)c_6 e^{-\kappa^2 y}$$

or

$$u(x, y) = A_2 e^{-\kappa^2 y} \cos 2\kappa x + B_2 e^{-\kappa^2 y} \sin 2\kappa x$$

Example

Find solutions of

$$\frac{\partial^2 u}{\partial x^2} = 4\frac{\partial u}{\partial y}.$$

3) Case 3:

$$\frac{X''}{4X} = \frac{Y'}{Y} = 0$$

Hence,

$$X'' = Y' = 0$$

which have the solutions

$$X(x) = c_7 x + c_8,\; Y(y) = c_9$$

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Example

Find solutions of

$$\frac{\partial^2 u}{\partial x^2} = 4\frac{\partial u}{\partial y}.$$

3) Case 3:

$$X(x) = c_7 x + c_8,\ \ Y(y) = c_9$$

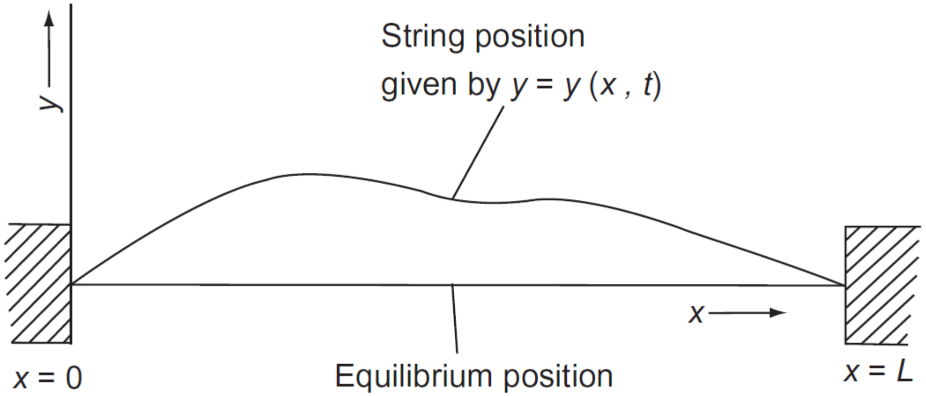
The final solution is

$$u(x, y) = X(x)Y(y) = (c_7 x + c_8)c_9$$

or

$$u(x, y) = A_3 x + B_3$$

Waves in a String



- Assumptions
  1. The string is perfectly flexible and homogeneous
  2. The displacements y are small compared to L
  3. It has a finite length and both ends are fixed

(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 12.6)

Wave Equation (파동방정식)

From the equation of motion (Newton's law),

$$\frac{\partial^2 y}{\partial t^2} = \frac{T}{\rho}\frac{\partial^2 y}{\partial x^2}$$

where T is the magnitude of the tension force and

ρ is the mass of the string per unit length.

Letting $c = \sqrt{T/\rho}$,

$$\frac{\partial^2 y}{\partial t^2} = c^2\frac{\partial^2 y}{\partial x^2}$$

Additional Conditions

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$$

- Boundary conditions

$$y(0, t) = y(L, t) = 0$$

- Initial conditions

$$y(x, 0) = 0, \qquad \left.\frac{\partial y}{\partial t}\right|_{t=0} = f(x)$$

Solving Wave Equation

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$$

Assume $y(x,t) = \psi(x)\theta(t)$;

$$\psi \frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} = c^2 \frac{\mathrm{d}^2\psi}{\mathrm{d}x^2}\theta$$

$$\underbrace{\frac{1}{c^2\theta}\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2}}_{\text{temporal part}} = \underbrace{\frac{1}{\psi}\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2}}_{\text{spatial part}} = \lambda$$

Spatial Part

$$\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2} = \lambda\psi$$

1. If $\lambda > 0$, we can set $\lambda = \kappa^2$ and obtain

$$\psi(x) = a_1 e^{\kappa x} + a_2 e^{-\kappa x}$$

2. If $\lambda = 0$, we obtain

$$\psi(x) = a_3 x + a_4$$

3. If $\lambda < 0$, we can set $\lambda = -\kappa^2$ and obtain

$$\psi(x) = a_5 \cos\kappa x + a_6 \sin\kappa x$$

Applying Boundary Conditions

$$y(0,t) = y(L,t) = 0 \ \rightarrow \ \psi(0) = \psi(L) = 0$$

1. If $\lambda > 0$, $\psi(x) = a_1 e^{\kappa x} + a_2 e^{-\kappa x}$

$$\psi(0) = a_1 + a_2 = 0,\ \psi(L) = a_1 e^{\kappa L} + a_2 e^{-\kappa L} = 0$$

$$\therefore a_1 = a_2 = 0$$

2. If $\lambda = 0$, $\psi(x) = a_3 x + a_4$

$$\psi(0) = a_4 = 0,\ \psi(L) = a_3 L + a_4 = 0$$

$$\therefore a_3 = a_4 = 0$$

Applying Boundary Conditions

$$y(0,t) = y(L,t) = 0 \;\rightarrow\; \psi(0) = \psi(L) = 0$$

3. If $\lambda < 0$, $\psi(x) = a_5 \cos \kappa x + a_6 \sin \kappa x$

$$\psi(0) = a_5 = 0,\; \psi(L) = a_5 \cos \kappa L + a_6 \sin \kappa L = 0$$

$$\therefore a_5 = 0 \text{ and } \kappa = n\pi/L$$

The only non-zero solution is

$$\psi(x) = a_6 \sin\left(\frac{n\pi x}{L}\right)$$

when $\lambda = -\kappa^2$ and $\kappa = n\pi/L$

Temporal Part

$$\frac{1}{c^2\theta}\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} = \frac{1}{\psi}\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2} = \lambda = -\kappa^2 = -\left(\frac{n\pi}{L}\right)^2$$

$$\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} = -\left(\frac{n\pi}{L}\right)^2 c^2\theta$$

Since the separation constant is negative,

$$\theta(t) = b_1\cos\left(\frac{n\pi c}{L}t\right) + b_2\sin\left(\frac{n\pi c}{L}t\right)$$

Applying Initial Conditions

$$\theta(t) = b_1 \cos\left(\frac{n\pi c}{L} t\right) + b_2 \sin\left(\frac{n\pi c}{L} t\right)$$

$$y(x, 0) = 0 \rightarrow \theta(0) = 0$$

Substitution leads to

$$\theta(0) = b_1 = 0$$

Hence,

$$\theta(t) = b_2 \sin\left(\frac{n\pi c}{L} t\right)$$

Total Solution

- Spatial solution

$$\psi(x) = a_6 \sin\left(\frac{n\pi x}{L}\right)$$

- Temporal solution

$$\theta(t) = b_2 \sin\left(\frac{n\pi c t}{L}\right)$$

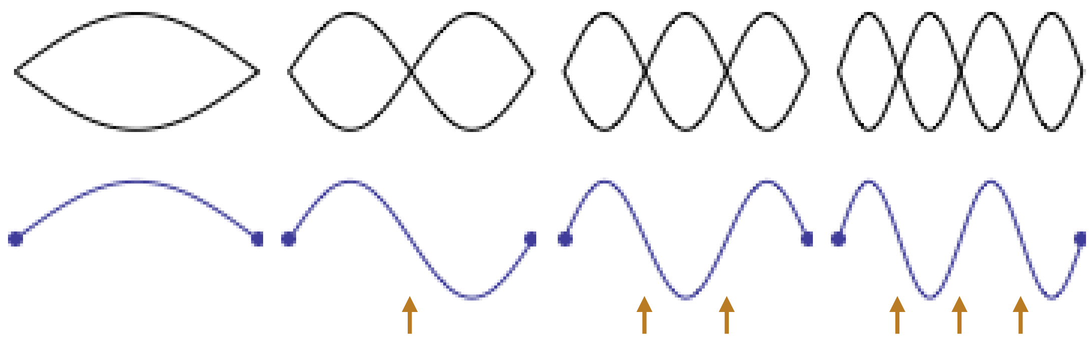
- Total solution

$$y(x,t) = \psi(x)\theta(t) = A \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{n\pi c t}{L}\right)$$

Standing Waves (정상파)

$$y(x,t) = \psi(x)\theta(t) = A\sin\left(\frac{n\pi x}{L}\right)\sin\left(\frac{n\pi ct}{L}\right)$$

$n = 1$ $n = 2$ $n = 3$ $n = 4$



$(n-1)$ nodes (마디)

(Image: https://www.acs.psu.edu/drussell/demos/standingwaves/standingwaves.html)

Traveling Waves (진행파)

$$y(x,t) = A\sin[k(x-ct)]$$

(Image: https://spacelights.tumblr.com/post/37429141540/infinity-imagined-standing-waves-a-standing-wave)

Example 12.12

Show that the function $y(x,t) = A\sin[k(x-ct)]$ satisfies the wave equation

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}.$$

The left-handed side is

$$\frac{\partial^2 y}{\partial t^2} = -Ak^2c^2\sin[k(x-ct)]$$

and the right-handed side is

$$c^2\frac{\partial^2 y}{\partial x^2} = -Ak^2c^2\sin[k(x-ct)]$$

Hence, the function satisfies the wave equation.

The Schrödinger Equation

- 1-dimensional time-dependent Schrödinger equation

$$i\hbar\frac{\partial\Psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\Psi}{\partial x^2} + \mathcal{V}(x,t)\Psi$$

  - $\Psi = \Psi(x,t)$: wave function
  - m: mass of a particle
  - $\mathcal{V}(x,t)$: potential energy
  - $\hbar = h/2\pi$, where h is Planck's constant
- Consider the case $\mathcal{V}(x,t) = \mathcal{V}(x)$!

Separation of Variables

$$i\hbar\frac{\partial\Psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\Psi}{\partial x^2} + \mathcal{V}(x)\Psi$$

Assume $\Psi(x,t) = \psi(x)\theta(t)$:

$$i\hbar\psi(x)\theta'(t) = -\frac{\hbar^2}{2m}\psi''(x)\theta(t) + \mathcal{V}(x)\psi(x)\theta(t)$$

Dividing both sides by $\psi(x)\theta(t)$,

$$\underbrace{i\hbar\frac{\theta'(t)}{\theta(t)}}_{\text{depends only on } t} = \underbrace{-\frac{\hbar^2}{2m}\frac{\psi''(x)}{\psi(x)} + \mathcal{V}(x)}_{\text{depends only on } x}$$

Separation Constant

$$i\hbar\frac{\theta'(t)}{\theta(t)} = -\frac{\hbar^2}{2m}\frac{\psi''(x)}{\psi(x)} + \mathcal{V}(x) = E$$

- Spatial part (time-independent Schrödinger eqn.)

$$-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2\psi}{\mathrm{d}x^2} + \mathcal{V}(x)\psi(x) = E\psi(x)$$

- Temporal part

$$i\hbar\frac{\mathrm{d}\theta}{\mathrm{d}t} = E\theta(t)$$

$$\theta(t) = Ce^{-iEt/\hbar}$$